LOT HW-001 ??? 01-BOM

HW-001 Bill of Materials and Buying List

Prices are estimates in USD at 100-unit volume, ±30%. Not quotes. Stock not verified. PCBWay turnkey needs manufacturer part numbers (MPNs), so MPNs are listed.

A. Prototype kit (buy now, Phase 1)

Item	Why	Link	Est.
Adafruit MEMENTO (ESP32-S3, 1.54” 240×240 TFT, camera, buttons)	Closest all-in-one reference	guide	$50
DFRobot FireBeetle 2 ESP32-S3 + OV2640 (DFR0975)	Cheaper OV2640 dev board	DigiKey	$25
Waveshare ESP32-S3-LCD-2	ST7789 screen + camera port	Waveshare	$20
BME280 breakout	Weather sensor	search “BME280 breakout” on Adafruit or SparkFun	$10
Qi receiver coil + module, 5 V	Test charging through window materials	search “Qi receiver 5V”	$10
Samples of 316L, glass, PEEK 40×40 plates	RF and Qi attenuation test	PCBWay CNC or local machinist	$60

B. Production BOM per unit (v1, 9.5 mm)

Function	Part (MPN)	Maker page	Est.
MCU + Wi-Fi/BLE	ESP32-S3-WROOM-1-N16R8 (pre-certified module)	Espressif	$5.00
Camera	OV2640 (OmniVision) module	OmniVision	$3.50
Screen	1.54” IPS 240×240, ST7789	vendor TBD (Waveshare/AliExpress/Adafruit)	$5.00
Weather sensor	BME280 (Bosch) T/RH/pressure	Bosch	$3.50
Upgrade: air quality “AI” sensor	BME688 (Bosch, runs BSEC)	Bosch	+$6
Wireless charging IC	BQ51013B (TI Qi receiver) + coil	TI	$4.50
Li-Po charger	MCP73831 or BQ25185	Microchip / TI	$0.80

Function	Part (MPN)	Maker page	Est.
Battery	Li-Po 301230–402025, 3.7 V, 100–150 mAh, with protection	vendor TBD	$3.50
3.3 V regulator	AP2112K-3.3	Diodes Inc.	$0.30
Button	Tactile switch, low profile (e.g. 3×2×0.6 mm)	TBD	$0.20
Piezo or LRA haptic	Pager alert	TBD	$0.80
PCB, 4-layer, 38×38 mm	PCBWay	PCBWay	$4.00
Assembly	PCBWay turnkey (quote-based)	quote	$10.00
Passives, connectors, FPC	misc.	—	$3.00
Charging window	PEEK or glass insert	—	$3.00
Steel body, 2 parts, 316L CNC	PCBWay CNC or machinist	—	$35–50
Mirror polish (one face)	finishing	—	$8–12
Gasket, screws, adhesive	—	—	$2.00
Packaging, QC, test jig share	—	—	$4.00
Total per unit			**about $105–125**

C. One-off costs (NRE)

Item	Est.
CNC programming and fixtures	$500–1,000
Stencil, test jig, programming fixture	$300
Pre-compliance RF scan	$1,000–2,000
FCC full test (only if selling)	$5,000–10,000
UN38.3 battery test (only if shipping commercially)	$1,500–3,000

100-unit run: about $10.5–12.5k hardware + about $1–2k NRE, before certification.

D. Cost levers

- Aluminium anodised or PVD-coated steel-look body cuts body cost by about 60%, but it is not stainless.
- Drop the camera: saves about $6 and gains about 4 mm of thickness.
- Order 300 units: the body drops to about $20–25 each.